GPU Waste Detection for Kubernetes: Market Research, Competition, Unit Economics & International Expansion

Scope: the product **as it is actually built today** (September 2026), not the broader vision. That product is a read-only, self-hosted Helm install that reads NVIDIA DCGM telemetry plus Kubernetes pod identity, classifies GPU waste into four categories (idle, over-provisioned, fragmented, recurring pattern), prices each finding in INR against a right-sizing ceiling, and shows it on a dashboard. It recommends; it does not act. It has been validated on GKE (L4), on a vast.ai/k3s GPU node, against the real Microsoft Philly training trace (16.7% of GPU-workload-hours flagged as waste) and the real Azure LLM inference trace (19,366/19,366 requests served, over-provisioning finding fired).

How to read the numbers: every figure is tagged **[S]** (sourced, see Sources at the end) or **[A]** (our assumption, listed in the Assumptions ledger). Market-research firm estimates vary widely and are treated as directional. Currency: INR; USD conversions at ₹88/USD [A].

Executive summary

Question	Answer
TAM	**Global:** ~$0.45B (2026) spent on cost visibility/optimization for GPU workloads on Kubernetes [S+A]. **India:** ~₹78-190 Cr/yr (~$9-22M) at this product's own price points [A], growing fast with India's GPU build-out.
SAM	**India, Kubernetes-based GPU fleets reachable with read-only, self-hosted, INR-priced software:** ~885 organizations, ~₹38 Cr/yr (~$4.4M) [A].
SOM (3 yrs)	**India only: 27-45 paying customers, ~₹0.8-1.35 Cr ARR (~$92-153k)** [A]. Viable as a bootstrapped business; not venture-scale on this product alone.
International	Adding SE Asia, Middle East, US and EU raises TAM to **~₹860 Cr (~$98M)** and SAM to **~₹270 Cr (~$31M)** [A]. With the same 45 customers (30 India + 15 abroad), year-3 ARR rises from ₹1.35 Cr to **~₹2.6 Cr** and net profit from ~₹62 L to **~₹1.13 Cr** [A]. Overseas customers pay ~3-5x more for similar support effort.
Verified re-estimate	Rebuilt top-down from published figures only (section 8): global TAM **~$312M (~₹2,980 Cr)**, SAM ~$250-287M; India TAM **~$5.9M (~₹56 Cr)**, SAM ~$4.7-5.4M. Lower than the bottom-up India TAM in section 2, which rests on assumed org counts and prices; prefer section 8 when citing figures externally.
Growing the market	Section 9 (Oct 2026): no source supports a ₹980 Cr/yr India market; plan against ~₹300 Cr/yr. Nine features widen it, led by automated actions and non-Kubernetes GPU support; realistic share even with all of them is 10-25% [A].
Earlier study	The Sept 2026 SMB market study's ₹216 Cr SAM / ₹7.2 Cr SOM sizes the **full planned product with a team**; with its price assumption corrected it is ₹130 Cr / ₹4.35 Cr. The gap to this report is the feature roadmap (section 7).
Main competitor	CAST AI ($1B valuation, Jan 2026) and ScaleOps ($800M+, Mar 2026) automate; Datadog launched GPU Monitoring (Apr 2026) and overlaps most directly on detection. You win only where read-only + self-hosted + INR + non-hyperscaler clusters matter.
Unit economics	Gross margin 78-89% (no hosting cost; you're self-hosted). LTV:CAC 3.4x (Starter) to 18x (Enterprise); CAC payback 4-6 months, founder-led [A]. As a solo founder, break-even is ~2 customers with no salary, ~5 with a ₹1 L/month draw [A]; your time (~25 customers supportable alone) is the real limit, not money.
Biggest risk	**One-time-audit churn:** a customer finds the waste, fixes it, cancels. If churn doubles, Starter LTV:CAC drops to 1.7x (unprofitable). Also: GPU-count pricing underprices H100 fleets and overprices cheap-GPU fleets.

1. Market context

Signal	Figure	Tag
India end-user public cloud spending (Gartner)	$17.5B in 2026, up 28.1% from $13.7B in 2025	[S]
India data-center GPU market (MarkNtel)	~$0.17B (2025) to $5.29B (2032), 63% CAGR	[S]
IndiaAI Mission compute	38,000+ GPUs onboarded at ~₹65/GPU-hr; target 100,000 by Dec 2026	[S]
Indian GenAI startups (Nasscom)	890+ in H1 CY2025 (740+ application layer); $990M cumulative funding	[S]
India GPU installed base (third-party survey)	80,000+ GPUs across public/private deployments	[S, 3rd-party blog]
Global GPU-as-a-Service (MarketsandMarkets)	$8.21B (2025) to $26.62B (2030), 26.5% CAGR	[S]
Global Kubernetes cost management market	~$1.75B (2025) to ~$2.23B (2026), ~27% CAGR; other estimates $1.38B (2025)	[S, analyst range]
Kubernetes for AI inference (CNCF 2025 survey)	66% of orgs hosting GenAI use Kubernetes for some or all inference	[S]
GPU share of compute cost (Datadog)	GPU instances = 14% of compute costs across its customer base	[S]

What this means: the value pool (GPU spend) is growing 25-60% a year, and Kubernetes is now the default place GPU inference runs. But the software pool this product can capture is a thin slice of GPU spend: cost-visibility tools typically capture 1% or less of the spend they watch.

2. TAM, SAM, SOM

2.1 Global TAM (top-down, for context)

Global Kubernetes cost-management spend is ~$2.23B in 2026 [S]. We assume 15-25% of it is GPU-driven [A], anchored on Datadog's finding that GPUs are already 14% of compute cost and rising [S], and on GPU nodes costing several times more per hour than CPU nodes. That gives **~$0.33-0.56B, midpoint ~$0.45B (~₹3,900 Cr)** of global spend on cost visibility and optimization for GPU workloads on Kubernetes. This is the category CAST AI, ScaleOps, Kubecost and Datadog are fighting over. It is **not** reachable by a two-person, INR-priced, founder-led product in the next three years; it matters as the ceiling and as the reason incumbents will keep investing here.

2.2 India TAM (bottom-up, at this product's own prices)

Segment (orgs running dedicated NVIDIA GPU fleets, 2026)	Count	Basis
AI / GenAI startups that train or self-host models	~900	[A] from Nasscom's 890+ GenAI startups [S]: most of the ~150 infra/model-layer firms, a minority of the 740 app-layer firms (most call APIs), plus non-GenAI AI firms (vision, speech, health AI)
Enterprises with in-house GPU clusters (BFSI, telecom, IT services, e-commerce, auto)	~600	[A]
Research, academic and government labs (IITs, IISc, IndiaAI compute users)	~300	[A]
GPU clouds / neoclouds / MSPs (E2E, Yotta, Neysa, NxtGen, Netweb, smaller)	~30	[A] better treated as channel partners than end customers
Total	**~1,800**	

Tier (current pricing)	GPUs	Annual price	Assumed share of orgs [A]
Starter	1-5	₹1.2 L	45%
Growth	6-25	₹3.6 L	35%
Scale	26-100	₹9.0 L	15%
Enterprise (assumed ₹2 L/mo)	100+	₹24 L	5%
Blended ACV at list		**₹4.35 L**	

Method A (by organization): 1,800 orgs x ₹4.35 L = **~₹78 Cr/yr (~$8.9M)**. **Method B (by GPU):** 80,000 GPUs [S, third-party] x ~₹2,000 per GPU-month (the Growth tier's effective per-GPU price at ~15 GPUs) x 12 = **~₹192 Cr/yr (~$21.8M)**. India TAM is therefore **~₹78-190 Cr/yr**. Method B is higher because it counts GPU-cloud and IndiaAI capacity that sits in a few very large fleets. With IndiaAI alone targeting 100,000 GPUs by Dec 2026 [S], this number should roughly double by 2027-28.

2.3 SAM (India, serviceable by the product as built)

The product requires Kubernetes, NVIDIA GPUs with DCGM, and a buyer willing to install a Helm chart. Filters applied [A]:

- Exclude GPU clouds as direct customers (channel instead): 1,770 end-user orgs.
- Keep only fleets run on Kubernetes: 50% [A]. This is deliberately below CNCF's 66% [S] because many Indian startups run GPUs on bare VMs, Slurm, or notebooks rather than Kubernetes.
- Result: **~885 orgs x ₹4.35 L = ~₹38.5 Cr/yr (~$4.4M)**.

Not filtered out, but harder to win inside the SAM: teams on EKS/GKE/AKS that already have CAST AI, Datadog, or hyperscaler cost tools, and teams whose GPU spend is subsidized (IndiaAI at ~₹65/GPU-hr shrinks the rupee value of the same waste roughly 4-5x vs. commercial A100 rates).

2.4 SOM (3-year obtainable)

Scenario	Share of SAM	Paying customers	Realized ACV [A]	ARR
Conservative	3%	27	₹3.0 L	₹0.81 Cr (~$92k)
Base	4%	35	₹3.0 L	₹1.05 Cr (~$119k)
Stretch	5%	45	₹3.0 L	₹1.35 Cr (~$153k)

Realized ACV is below the ₹4.35 L list blend because early customers skew small and pilots get discounted [A]. 3-5% of SAM in three years is normal for a founder-led product with no sales team.

Honest read: the product as built supports a profitable small business (break-even for a solo founder is 2-5 customers, section 4), not a venture-scale one. Getting to venture scale needs one or more of: (1) selling outside India (SE Asia, Middle East, US, EU: a ~$26M international SAM, see section 5); (2) white-labelling through Indian GPU clouds to reach their customers; (3) moving from detection to action (the queue-depth/TTFT and bin-packing items from the MVP roadmap), which is what justifies CAST AI's and ScaleOps's valuations.

3. Competitive landscape

Four kinds of competitor: **automation platforms** that change your cluster (CAST AI, ScaleOps, PerfectScale), **cost-allocation tools** (Kubecost/OpenCost), **observability platforms** adding GPU views (Datadog), and **schedulers** (NVIDIA Run:ai / KAI). The default competitor is still **DIY**: dcgm-exporter + Prometheus + Grafana.

Competitor	What it is	GPU capability	Pricing (public / reported)	Scale	Overlap
CAST AI	Kubernetes automation platform (autoscaling, bin-packing, Spot, rightsizing)	GPU node autoscaling, Spot GPUs, time-slicing/MIG sharing, GPU cost monitoring, OMNI Compute / GPU marketplace	Free monitoring; paid tiers reported at $200 / $1,000 / $5,000 per month; savings-share ~15-20% reported (third-party)	$1B valuation (Jan 2026); ~$181M raised	High on visibility; they act, you recommend
ScaleOps	Autonomous real-time Kubernetes resource management	Adjusts compute and GPU resources automatically	Not public (enterprise)	$130M Series C at $800M+ (Mar 2026); $210M total; 350%+ YoY growth	Medium-high
Datadog GPU Monitoring	Observability platform add-on (GA Apr 22, 2026)	Fleet health, idle/misconfigured GPUs, cost + chargeback by team, cloud / neocloud / on-prem	Usage-based on top of Datadog agent pricing	Public company; huge installed base	**Highest on detection**
IBM Kubecost / OpenCost	Kubernetes cost allocation (acquired by IBM Sep 2024; OpenCost is the CNCF open-source core)	Per-GPU hourly cost allocation; idle-capacity cost; not GPU-utilization waste classification	Free/OSS tier; Business plan reported ~$449/mo per 100 nodes	IBM FinOps suite (Apptio, Cloudability, Turbonomic)	Medium
PerfectScale (DoiT)	Kubernetes optimization and governance (acquired by DoiT Feb 2025)	Rightsizing across CPU, GPU, storage, networking	Bundled with DoiT services	Part of DoiT	Medium
NVIDIA Run:ai / KAI Scheduler	GPU scheduler/orchestrator; KAI is the Apache 2.0 open-source core (CNCF Sandbox)	Fair-share quotas, gang scheduling, GPU sharing; utilization dashboards in Run:ai	KAI free; Run:ai enterprise	NVIDIA	Low (different layer)
DIY: DCGM + Prometheus + Grafana	Free open-source metrics stack	Raw utilization per GPU; no waste taxonomy, no workload rollups, no ₹ recommendations out of the box	Free (engineer time)	Everywhere	High on raw data
Indian GPU clouds' own consoles	E2E, Yotta, etc. instance dashboards	Basic per-instance utilization for their own GPUs	Included	Growing fast	Low; potential channel partners

3.1 Feature comparison vs. CAST AI (closest well-known name)

Capability	Your product	CAST AI
GPU waste detection (idle / over-provisioned / fragmented / recurring)	Yes, four categories, confidence-scored	Partial (utilization + cost monitoring; not this taxonomy)
₹ savings estimate per finding	Yes, native INR, SLA-aware right-sizing ceiling	USD cost; savings reported after automation
Automatic action (scale, rebin, Spot, sharing)	**No** (recommends only)	**Yes** (core product)
Cluster permissions	**Read-only** (get/list/watch only)	Needs write access for automation
Where data lives	**Customer's own cluster** (self-hosted, SQLite)	CAST AI SaaS
Cluster types	Any Kubernetes with NVIDIA+DCGM; tested GKE and k3s/vast.ai	Strongest on EKS/GKE/AKS
Cloud-billing integration	No	Yes
Serving metrics (queue depth, TTFT)	No (roadmap)	Partial
Multi-cluster, SSO, SOC 2	No	Yes
Pricing model	Flat INR subscription by GPU count	USD tiers + savings share

3.2 Where you win and where you lose

You win when...	You lose when...
- Security won't grant write access to a third-party agent (banks, government, regulated health) - Data must stay in-cluster or in-country (DPDP Act, sovereign-AI projects) - The cluster is on-prem, a neocloud, or k3s, not EKS/GKE/AKS - The buyer wants a flat INR invoice, not a USD savings share - The team is small (5-50 GPUs) and not a Datadog customer	- The buyer wants savings to happen automatically (CAST AI, ScaleOps) - They already pay Datadog and can switch on GPU Monitoring - They need billing reconciliation, multi-cluster, SSO, SOC 2 - Their GPUs are subsidized and the waste isn't worth ₹ to them - An engineer says "we'll build it in Grafana"

Defensibility is low today. The detection logic can be copied, and Datadog shipped a similar capability this year. Your advantages are distribution (India, INR, founder-led), the trust model (read-only, self-hosted), and speed. The durable moat has to come later from real pilot data (calibrated thresholds, per-workload baselines) and from being the tool Indian GPU clouds bundle.

4. Unit economics

The product is self-hosted in the customer's cluster, so you pay nothing for compute or storage per customer. Cost of goods is mostly **your support time**. All inputs below are assumptions [A] until pilots produce real data; founder time is costed at ₹1,500/hour [A].

4.1 Per-customer, monthly

Line item	Starter	Growth	Scale	Enterprise
Price / month	₹9,999	₹29,999	₹74,999	₹2,00,000 [A]
Hosting / compute	₹0	₹0	₹0	₹0
Shared tooling (CI, registry, email, domain)	₹500	₹500	₹500	₹500
Support hours / month [A]	1 h = ₹1,500	2 h = ₹3,000	5 h = ₹7,500	12 h = ₹18,000
Payment processing (2%) [A]	₹200	₹600	₹1,500	₹4,000
COGS	₹2,200	₹4,100	₹9,500	₹22,500
Gross profit / month	**₹7,799 (78%)**	**₹25,899 (86%)**	**₹65,499 (87%)**	**₹1,77,500 (89%)**

4.2 Acquisition, lifetime value, payback

Metric	Starter	Growth	Scale	Enterprise
CAC: founder-led sales, incl. ~4 free scans per win [A]	₹40,000	₹90,000	₹2,50,000	₹6,00,000
Onboarding (install, 7-day scan, readout) [A]	4 h = ₹6,000	8 h = ₹12,000	16 h = ₹24,000	40 h = ₹60,000
Total acquisition cost	₹46,000	₹1,02,000	₹2,74,000	₹6,60,000
Monthly churn [A]	5%	3%	2%	1.5%
Expected lifetime	20 months	33 months	50 months	67 months
LTV (gross profit x lifetime)	₹1.56 L	₹8.63 L	₹32.7 L	₹1.18 Cr
LTV : CAC	**3.4x**	**8.5x**	**12.0x**	**17.9x**
CAC payback	5.9 months	3.9 months	4.2 months	3.7 months
LTV : CAC if churn doubles	1.7x	4.2x	6.0x	9.0x

Rule of thumb: LTV:CAC above 3x and payback under 12 months is healthy. Every tier clears that on paper, but **Starter is marginal**: it breaks if churn is higher than assumed, and it breaks as soon as a paid salesperson replaces founder time. CAC is founder-time CAC; hiring an account executive at ~₹1.5-2 L/month [A] roughly doubles to triples CAC for the lower tiers.

4.3 Customer-side ROI: does the price make sense to the buyer?

Recoverable waste assumed at **10% of GPU spend** [A]. Basis: the Philly replay flagged 16.7% of GPU-workload-hours as waste [S, own test; 14.7% excluding unverified consolidation candidates]; not all flagged waste is recoverable (SLA headroom, team inertia), so if 30-50% of it gets acted on that is ~5-8%. **10% is therefore at the optimistic end**; at 7%, every ROI below is 30% lower (e.g. 16 x A100: 8.2x instead of 11.7x; 4 x L4: 1.4x). GPU rates are the product's own illustrative INR defaults (L4 ₹70/h, A100 80GB ₹300/h, H100 SXM ₹520/h), and IndiaAI's ~₹65/h [S]; 730 hours/month.

Customer fleet	GPU spend / month	Tier	Price as % of spend	10% recovered	ROI (x price)
4 x L4	₹2.04 L	Starter	4.9%	₹20,440	**2.0x** (weak)
16 x GPUs on IndiaAI subsidy	₹7.59 L	Growth	4.0%	₹75,920	**2.5x** (weak)
4 x A100 80GB	₹8.76 L	Starter	1.1%	₹87,600	**8.8x**
16 x A100 80GB	₹35.0 L	Growth	0.86%	₹3.50 L	**11.7x**
64 x H100 SXM	₹2.43 Cr	Scale	0.31%	₹24.3 L	**32x** (underpriced)

Pricing problem: pricing by GPU count ignores that an H100 costs ~7x an L4. Small fleets of cheap or subsidized GPUs get a weak 2-2.5x return (a 2x ROI is usually not enough for a buyer to bother), while large H100 fleets pay 0.3% of spend for a 32x return. **Recommendation:** price on monitored GPU spend, e.g. ~1% of monitored monthly GPU spend with a ₹9,999 floor and volume discounts [A]. That keeps ROI around 10x for everyone and grows revenue with the customer's fleet.

4.4 Company break-even (solo founder)

You are a solo founder doing engineering and sales, and the product has no hosting cost, so the company's fixed costs are small [A]:

Monthly fixed cost	No salary	With ₹1 L/month personal draw
GPU test rentals (occasional)	₹10-20k	₹10-20k
CA / GST compliance	₹5-10k	₹5-10k
Tools, domain, email	₹3-5k	₹3-5k
Travel to customer meetings	₹10-20k	₹10-20k
Founder draw	₹0	₹1,00,000
Total (rounded)	**~₹50k**	**~₹1.5 L**
Break-even customers (avg ~₹31,300/month gross profit at the 45/35/15/5 tier mix)	**~2**	**~5**

Your time is the constraint, not money. At ~2.5 support hours per customer per month and ~40% of a 160-hour month on support, one person can look after **~25 customers**. Winning one customer takes ~30 hours (about 4 free scans plus onboarding), so ~30 sales hours a month gives ~1 new customer a month, roughly the 3-year SOM. Two levers stretch this: make the free scan self-serve (cuts a scan from ~6 hours to ~1), and favour Growth tier and above. **First hire** (a support engineer, ~₹1.25 L/month): when support passes ~40% of your time or ~20-25 customers (~₹60-75 L ARR).

4.5 What would change these numbers most

- **Churn** (the biggest swing factor). If customers treat the product as a one-time audit, lifetime collapses. Counter-measures: continuous value (weekly ₹ reports, regression alerts when waste returns, per-team chargeback) and annual contracts.
- **Free-scan conversion.** CAC assumes ~1 in 4 free scans converts. At 1 in 10, lower-tier CAC more than doubles.
- **Pilot-proven savings.** The whole ROI table rests on the 10% recoverable assumption. The first pilot's before/after number replaces it.
- **Moving to action** (bin-packing recommendations, queue-depth-aware right-sizing) supports a higher price and lower churn, but brings you into direct competition with CAST AI and ScaleOps.

5. International expansion

Overseas buyers tend to run larger GPU fleets and pay in USD, so each customer is worth ~2-6x an Indian one for broadly similar support effort. That matters most for a solo founder, whose limit is customer count, not money. Four candidate regions:

Region	Market signals [S]	Competition	Fit for this product
SE Asia (Singapore, Malaysia, Indonesia, Philippines)	680+ AI startups, $2.3B AI funding in 12 months (e-Conomy SEA 2025); Malaysia leads SE Asia data-centre construction (1,039 MW); Aolani + NVIDIA plan 22,000 GPUs in Malaysia/Philippines	Lighter; global tools present but little local focus	**Good.** Close time zones, similar price sensitivity, new GPU clouds as channel partners
Middle East (UAE, Saudi Arabia)	MENA AI startups raised $858M in 2025 (60% UAE); 673 AI companies in Abu Dhabi; HUMAIN ordered 18,000 GB300 GPUs; G42's Stargate UAE first stage ~400,000 NVIDIA chips	Sovereign buyers work through large integrators; few SaaS tools allowed in	**Good for the trust model.** Read-only + self-hosted suits sovereign data rules, but needs a local partner; the mega-fleets are enterprise deals out of solo reach
United States	~6,956 AI startups (largest share globally); neocloud revenue >$25B in 2025, projected ~$42B in 2026	**Heaviest:** CAST AI, ScaleOps, Datadog, Kubecost all sell here	**Largest pool, hardest sale.** Target neocloud/on-prem GPU users, not EKS/GKE teams; SOC 2 required
EU + UK	~1,192 GenAI companies in Europe; UK 800+ AI startups, France 468, Germany 394	CAST AI (EU-founded), Datadog, Kubecost	**Moderate.** Data-residency rules favour self-hosted; GDPR paperwork; SOC 2 / ISO 27001 expected

5.1 TAM, SAM, SOM by region (bottom-up, at this product's prices)

Pricing assumed at ~1% of monitored GPU spend in USD. Org counts and average contract values are our assumptions [A], anchored on the startup and funding counts above [S]. SAM keeps orgs on Kubernetes (CNCF: 66% of GenAI hosts [S]) and, in the US and EU, only the share that values a read-only, self-hosted tool or runs on neocloud/on-prem clusters.

Region	Orgs with GPU fleets [A]	Avg contract / yr [A]	TAM	SAM filter [A]	SAM	3-yr SOM [A]
India	~1,800	₹3-4.35 L	₹78 Cr ($8.9M)	50% on K8s	₹38 Cr ($4.4M)	30 cust, ₹0.90 Cr
SE Asia	~370	$8k (₹7 L)	₹26 Cr ($3.0M)	55% on K8s	₹14 Cr ($1.6M)	6 cust, ₹0.42 Cr
Middle East	~250	$20k (₹17.6 L)	₹44 Cr ($5.0M)	55% on K8s	₹24 Cr ($2.8M)	3 cust, ₹0.53 Cr
United States	~4,200	$15k (₹13.2 L)	₹554 Cr ($63M)	66% K8s x 40% fit	₹146 Cr ($16.6M)	4 cust, ₹0.53 Cr
EU + UK	~1,500	$12k (₹10.6 L)	₹158 Cr ($18M)	60% K8s x 50% fit	₹48 Cr ($5.4M)	2 cust, ₹0.21 Cr
Intl. only	~6,300		**₹783 Cr ($89M)**		**₹233 Cr ($26.4M)**	**15 cust, ₹1.69 Cr**
India + intl.	~8,100		**~₹860 Cr ($98M)**		**~₹270 Cr ($31M)**	**45 cust, ~₹2.6 Cr**

Consistency check: the bottom-up total (~$98M) is about 22% of the ~$446M global top-down TAM in section 2.1. The top-down figure also counts very large hyperscaler customers, automation platforms' revenue and spend this product can't serve, so a bottom-up number well below it is expected. **Why SOM is small abroad:** 15 overseas customers is 0.4-3% of each region's SAM, limited by one founder's selling time, not by demand.

6. Comparison: India only vs. India + international (year 3)

Same total of 45 customers in both cases, since one founder plus one or two support engineers can't handle many more. Figures in ₹ lakh per year; all costs are assumptions [A].

	India only (45 India)	**India + international (30 India + 15 abroad)**
Revenue (ARR)	**135**	**259**
Payment processing (+ FX abroad)	2.7 (2%)	6.5 (2.5%)
CI, registry, tooling	1.0	1.5
Support engineers	15 (one)	30 (two; one covering US/EU hours)
Gross profit	**116 (86%)**	**221 (85%)**
SOC 2 / security certification	4	20 (required by overseas buyers)
US company + US accountant	0	3
Travel	2.4	12 (2-3 overseas trips)
Conferences (attend, not exhibit)	0	5
CA, legal, tools, GPU testing	3.6	6
Profit before your salary	**106 (79%)**	**175 (68%)**
Your salary (₹2 L/month)	24	24
Corporate tax (~25%)	21	38
Net profit	**62 (46%)**	**113 (44%)**

	India only	**India + international**
TAM / SAM	₹78 Cr / ₹38 Cr	~₹860 Cr / ~₹270 Cr
Avg revenue per customer	₹3 L	₹5.8 L
Competition	Light; INR pricing and local presence are real advantages	Heavy in the US; your edge narrows to read-only / self-hosted / neocloud
Fixed costs to enter	Minimal	SOC 2, US entity, travel, conferences, 2nd support engineer: ~₹50 L/yr extra
Sales cycle and trust	Shorter; in-person possible	Longer; security reviews; solo-founder risk questions
Can you stay solo?	Until ~25 customers	Needs a support hire earlier because of time zones
Risk if it goes badly	Low: costs stay small	Higher: ~₹50 L/yr of fixed costs before overseas revenue

Recommendation: sequence it, don't choose. (1) Now to year 1: India only. Get pilot-proven savings and 2-3 written case studies; you need them anywhere. (2) Years 1-2: SE Asia and the Middle East, sold remotely and through GPU-cloud partners; close time zones and a trust model that suits sovereign buyers. (3) Years 2-3: US and EU, once SOC 2 and case studies exist, targeting neocloud and on-prem GPU users rather than EKS/GKE teams already on CAST AI. This keeps the ~₹50 L/yr of international fixed costs until revenue can pay for them.

7. Reconciling with the earlier market study

An earlier study, India GPU Optimization SaaS: SMB Market Research (September 2026), sized the same opportunity much larger. Both are planning estimates built on assumptions; the difference is mostly **what product is being sized** and **who is selling it**, not a disagreement about the market.

	Earlier study (base case)	Earlier study, price corrected [A]	This report (India, as built)
Product sized	Full plan: Kubernetes + GPU VMs, billing, approve-then-act fixes, verified savings, multi-cluster	Same full plan	Kubernetes only; detect and recommend; read-only
Seller	A team	A team	One founder
Orgs (TAM / SAM)	12,000 / 3,000	12,000 / 3,000	1,800 / 885
Avg contract / yr	₹7.2 L (₹60k/month)	₹4.35 L	₹4.35 L list, ₹3 L realized
TAM	₹720 Cr (range 250-1,750)	₹522 Cr	₹78-190 Cr
SAM	₹216 Cr (range 60-600)	₹130 Cr	₹38 Cr
SOM, year 3	100 customers, ₹7.2 Cr (range 1.2-30)	100 customers, ₹4.35 Cr	27-45 customers, ₹0.8-1.35 Cr

7.1 Why the numbers differ

- **Product scope.** The earlier study includes teams running GPUs on plain VMs without Kubernetes, which the current product can't serve. That alone roughly halves the reachable market [A].
- **Organisation count.** 12,000 vs ~1,800. Neither is sourced; the earlier study says so itself. 12,000 likely includes occasional single-GPU renters who won't pay for optimisation; 1,800 counts dedicated fleets built up from Nasscom's ~890 GenAI startups plus enterprises and labs. The real number is probably between.
- **Average price.** ₹60k/month implies most customers on the 26-100 GPU Scale tier, which contradicts the study's own 1-50 GPU SMB target and its ₹9,999 / ₹29,999 / ₹74,999 tiers. At the tier mix used here the blended price is ₹4.35 L/yr; that correction alone cuts the earlier TAM and SAM by ~28%.
- **Seller capacity.** 100 customers in 3 years is ~4x what one founder can support (~25). The earlier study's upside (300 accounts, ₹30 Cr) needs a sales and support team. At the low end the two agree: its conservative ₹1.2 Cr SOM sits between this report's base (₹1.05 Cr) and stretch (₹1.35 Cr).

7.2 The gap is the roadmap

Each missing feature widens the market the product can serve or raises what each customer pays. Effects are rough assumptions [A] to size priorities, not forecasts.

Feature (in earlier study, not built)	What it unlocks	Effect on market / price [A]	When
Reuse an existing dcgm-exporter (Helm value)	Neocloud and GPU-Operator clusters that already run one	Removes an install blocker; no new segment	Now (hours)
GPU VM support (no Kubernetes)	Teams renting bare GPU VMs, common on Indian GPU clouds	SAM roughly 2x (₹38 Cr to ~₹75 Cr)	If pilots show demand
Cloud / GPU-cloud billing integration	Real invoice costs instead of rate tables; before/after savings proof	Enables a verified-savings fee (10-15% of savings); lowers churn	After first pilots
Savings verification (baseline vs after)	Proof the fixes worked	Turns the 10% assumption into a measured number; supports 2% pricing	With billing
Approve-then-act fixes (Slack/Jira, rollback, audit)	Engineers approve a change, product applies it	~1.5x price; less one-time-audit churn	Year 1-2
Serving metrics (queue depth, TTFT, cost per token)	Inference teams that care about latency	Stronger fit for inference buyers; safer right-sizing	Year 1-2
Multi-cluster, SSO, audit export	Scale and Enterprise tiers	Unlocks ₹75k+/month deals	Year 2

How to use the two documents. Use this report to plan the business you are running now (costs, break-even, when to hire). Use the earlier study's framing to describe the long-term vision to investors or partners, with the corrected ₹4.35 L average price and its organisation counts labelled as assumptions. The earlier study's positioning, free-scan offer, GPU-cloud partnerships and validation plan (20 interviews, 5 read-only pilots, 2 paid pilots with verified 15-25% savings) remain the right next steps.

8. Independent, source-verified market sizing (October 2026)

A from-scratch re-estimate that uses only published figures and none of this report's earlier assumptions. It sizes the product exactly as built: self-hosted, read-only waste detection and ₹ estimates for **NVIDIA GPUs on Kubernetes**, recommending without acting. Where no published figure exists, the assumption and the arithmetic are shown. Several research-publisher sites could not be opened directly; a figure is used only where multiple independent outlets report it identically, and the table says which. Converted at ₹95.45/$ (September 2026 average) [S].

8.1 Verified inputs

Fact	Figure	How verified
Kubernetes cost management market	$1.38B (2024), $1.75B (2025), **$2.23B (2026)**	Two publishers independently give $1.38B for 2024 (The Business Research Company, Growth Market Reports); TBRC 2025/2026 figures carried by several outlets
GPU monitoring software market	**$1.28B (2024)**, 13.6% CAGR	Single publisher (Growth Market Reports); primary page not reachable -- weaker
GPU share of compute costs	**14%**	Datadog's own GPU Monitoring press release, 22 Apr 2026 (Datadog AWS customer base)
Organizations running GPU workloads	**6%** of organizations (Oct 2025)	Datadog State of Containers and Serverless page; skewed toward heavy cloud users
Kubernetes for GenAI inference	66% of orgs hosting GenAI use it for some or all inference	CNCF 2025 Annual Cloud Native Survey (widely reported)
NVIDIA share of data-center GPUs	80-92% depending on measure	Multiple analyst reports
Public cloud spend, worldwide 2025	**$723.4B**	Gartner, consistently reported
Public cloud spend, India	**$13.7B (2025)**, $17.5B (2026)	Gartner press release, 1 Jun 2026
Indian companies using Kubernetes	**2,700 to 28,674** (databases disagree ~10x)	TheirStack; ELP Data
USD/INR	₹95.45 (Sept 2026 average)	Trading Economics

8.2 Global

Level	How it's calculated	Size (2026)
Market (GPU monitoring software, closest category)	$1.28B (2024) grown 13.6% for two years	**~$1.65B (~₹15,800 Cr)**
TAM (cost tooling for GPUs on Kubernetes)	$2.23B K8s cost management x 14% GPU share of compute cost	**~$312M (~₹2,980 Cr)**
SAM (NVIDIA only, as built)	TAM x 80-92% NVIDIA share	**~$250-287M (~₹2,390-2,740 Cr)**
SOM (3 years) [A]	0.1-0.5% of SAM (see 8.5)	**~$0.25-1.4M ARR (~₹2.4-13.7 Cr)**

8.3 India

Level	How it's calculated	Size (2026)
Market	Global market x India's 1.89% share of public cloud spend (13.7 / 723.4, Gartner 2025)	**~$31M (~₹299 Cr)**
TAM	Global TAM x 1.89%	**~$5.9M (~₹56 Cr)**
SAM	TAM x 80-92% NVIDIA	**~$4.7-5.4M (~₹45-52 Cr)**
SOM (3 years) [A]	2-5% of SAM (see 8.5)	**~$95-270K ARR (~₹0.9-2.6 Cr)**

Bottom-up cross-check: 2,700-28,674 Indian Kubernetes companies x 6% running GPUs = ~160-1,700 organizations. Spread over the India SAM that is ~$2.7k-34k per organization per year, a plausible tool budget, so the top-down and bottom-up views are consistent.

8.4 Comparable valuations [S]

- **Cast AI:** valued at over $1B, January 2026 (Cast AI press release).
- **ScaleOps:** over $800M, $130M Series C, March 2026 (Insight Partners).
- Both automate changes and sell mostly outside India; the product as built does neither.

8.5 What is assumption, not fact

- No source measures spend on GPU-cost tools for Kubernetes specifically; the TAM assumes tool spend follows GPUs' 14% share of compute cost.
- India is approximated by its share of public cloud spend. India's GPU capacity is growing faster than cloud overall (IndiaAI: 38,000+ GPUs onboarded, 100,000 targeted by Dec 2026), so the India figures are probably understated.
- SOM cannot be sourced; it is a capture-rate assumption. Only benchmark found: Kubecost, ~$6.2M revenue (third-party estimate) against the ~$1.38B 2024 market (~0.45%) after ~5 years and $30.5M raised before IBM acquired it (Sept 2024). A solo founder over 3 years should expect the low end globally; a home market supports a higher share, hence 2-5% for India.
- Market-research figures are single-firm, unaudited estimates: order-of-magnitude only.

Bottom line: the market is real and growing fast (Kubernetes cost management ~27% from 2025 to 2026; GPU-instance spend at Datadog's customers +40% in a year), but a read-only, NVIDIA-only, Kubernetes-only product addresses a small slice: ~$300M globally and ~$5-6M in India today. That supports a profitable small business from India, and venture scale only by selling globally or widening what the product does. Note: section 8 uses ₹95.45/$; earlier sections use ₹88/$.

9. Growing the market: features to build later (verified, October 2026)

Which features would let the product address the broader Indian market for GPU cost optimization, not just the read-only, NVIDIA-on-Kubernetes slice sized in section 8. Each claim is tagged: **[S]** checked against a published source; **[R]** reported in the press, primary source not opened; **[A]** assumption.

9.1 How big the target really is

- **A ₹980 Cr/yr India market for GPU cost optimization is not supported by any source found.** No published estimate of that size exists; the nearest figure in this report is the global TAM of ~₹2,980 Cr (section 8.2).
- India's whole GPU-monitoring software market is **~₹299 Cr** (section 8.3: global market x India's share of cloud spend) [S+A].
- India's entire cloud-GPU services market: **$67.31M (~₹640 Cr) in FY2025, rising to $513.67M by FY2030** (50.15% CAGR), with average capacity utilisation **below 20%** per expert interviews -- figures from an IPO prospectus as reported [R]. A ₹980 Cr tools market would exceed all the cloud-GPU rental it optimizes.
- **Plan against a ceiling of ~₹300 Cr/yr today**, growing quickly with India's GPU spend (public cloud +28% in 2026, IaaS +40% [S]).

9.2 Features that widen the market, in suggested build order

Feature	Why it matters (evidence)	What it adds
1. Automated actions (approve-then-act first): reclaim idle GPUs, scale down in quiet hours, apply right-sizing	CAST AI already automates idle GPU node termination, spot with on-demand fallback, MIG/time-slicing and rightsizing [S]; valued over $1B, ScaleOps over $800M (section 8.4) [S]. Detection-only competes directly with Datadog GPU Monitoring (GA Apr 2026) [S].	Higher price per customer: a tool that acts can charge more [A]
2. GPU VMs, bare metal and Slurm (no Kubernetes)	66% of organisations hosting GenAI use Kubernetes for "some or all" inference [S] -- about a third do not. E2E, India's largest GPU cloud, sells VMs, bare metal and managed Kubernetes [S]. India's PARAM supercomputers schedule GPU jobs with Slurm [S].	More customer organisations; roughly doubles India SAM [A]
3. GPU sharing: recommend and set up MIG / time-slicing	Indian cloud-GPU utilisation below 20% [R]. CAST AI and NVIDIA's KAI Scheduler both provide GPU sharing [S].	Larger savings per customer
4. Inference-serving optimisation: vLLM/Triton metrics, batching, autoscaling on request load, model-to-GPU fit	Gartner: inference spending overtakes training in 2026, 55% of AI-optimised IaaS spend [S, via press coverage].	Covers the largest, fastest-growing category of GPU spend
5. Chargeback, budgets and forecasting by team / business unit	Datadog's product chief: companies "can't chargeback GPU spend across business units" [S]; Datadog forecasts buy-more-GPUs vs optimise [S]; Kubecost sells chargeback as an enterprise feature [S].	Finance/CFO budgets; multi-team enterprises
6. Billing and purchasing advice: real invoice cost, spot, reserved/committed GPUs, moving eligible work to IndiaAI-subsidised compute	IndiaAI: 38,000+ GPUs onboarded at subsidised rates (sources quote ₹65 or ₹115-150 per GPU-hour), target 100,000 by Dec 2026 [S].	An India-only edge no global tool has [A]

Feature	Why it matters (evidence)	What it adds
7. Non-NVIDIA accelerators: Google TPU, AWS Trainium/Inferentia, AMD	NVIDIA holds 80-92% (section 8.1) [S]; IndiaAI's third tender adds 1,050 Google Trillium TPUs, its first non-NVIDIA chips [S].	The remaining 8-20% of accelerators
8. Multi-cluster, SSO, RBAC, audit logs	Kubecost puts SSO, RBAC and multi-cluster in its enterprise tier [S].	Required to sell to large enterprises
9. Scheduling, quotas and queues, built on NVIDIA's open-source KAI Scheduler	KAI Scheduler open-sourced under Apache 2.0 in April 2025, now a CNCF project [S].	Platform-team budgets
Distribution (not a feature): partner with Indian GPU clouds	Yotta, E2E, NxtGen and Jio lead the IndiaAI compute-provider roster [S].	Likely the fastest route to many Indian customers [A]

9.3 Realistic share, even with every feature

- No one captures the whole market. Closest benchmark: Kubecost reached ~0.45% of its global market after ~5 years (third-party revenue estimate, section 8.5) [R].
- A strong India leader in its home market might reach **10-25%** [A]: **~₹30-75 Cr/yr** of a ~₹300 Cr market.
- Features 1-9 are several years of work; realistically a team of ~15-40 people [A].
- Build each only after pilots show customers pay for it; the first paid pilot decides whether 1 or 2 comes first.

10. Assumptions ledger

#	Assumption	Value	Why / how to validate
A1	USD/INR	₹88	Approximate 2025-26 rate
A2	GPU share of Kubernetes cost-management spend	15-25%	Datadog: GPUs 14% of compute cost and rising; GPU nodes cost more per hour
A3	Indian orgs running dedicated NVIDIA GPU fleets	~1,800	Built from Nasscom startup counts + estimates; validate via GPU-cloud partner conversations
A4	Share of those on Kubernetes	50%	Below CNCF's 66%; ask every prospect
A5	Tier mix of customers	45/35/15/5	Skewed small; replace with pipeline data
A6	Enterprise price	₹2 L/month	Custom tier; placeholder
A7	Realized ACV	₹3.0 L	Discounts + small-customer skew
A8	Founder time value	₹1,500/hour	Opportunity cost placeholder
A9	Support / onboarding hours	See 4.1-4.2	Measure on first pilots
A10	CAC per tier	₹40k to ₹6 L	Founder-led, ~4 free scans per win
A11	Monthly churn	1.5-5%	Biggest uncertainty; see 4.5
A12	Recoverable waste	10% of GPU spend	Optimistic: Philly 16.7% flagged x 30-50% acted on = 5-8%; prove in pilot
A13	Monthly fixed costs (solo founder)	₹50k-1.5 L	With or without a ₹1 L personal draw
A14	3-year SAM capture (India)	3-5%	Typical for founder-led B2B infrastructure software
A15	Overseas orgs with GPU fleets	SEA 370, ME 250, US 4,200, EU 1,500	Anchored on startup counts in section 5; validate with regional partners
A16	Overseas average contract	$8k-20k/yr	~1% of monitored GPU spend; larger fleets abroad
A17	Overseas SAM filters	55-66% on K8s; 40-50% fit in US/EU	Share that values read-only, self-hosted or runs neocloud/on-prem
A18	International entry costs	~₹50 L/yr	SOC 2, US entity, travel, conferences, 2nd support hire
A19	Solo support capacity	~25 customers	2.5 h/customer/month, 40% of time on support

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Market-research-firm figures differ substantially between publishers; treat them as ranges. Competitor pricing marked third-party was not confirmed on the vendor's own pricing page.